

Nathan Xia

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ABOUT

First year Engineering Science student with experience in CFD, CAD, and aerodynamic design through national science competitions and design teams. Interested in control systems, fluid control, and numerical simulations.

EDUCATION

- University of Toronto** Toronto, Canada
• *Bachelor of Applied Science - Engineering Science | Lachlan Dales McKeller Scholar* *September 2025 - Present*
cGPA: 3.95/4.00
- Henry Wise Wood High School** Calgary, Canada
• *Course Average: 98.0%; IB Math Analysis and Approaches HL, IB Physics HL* *September 2022 - June 2025*

SKILLS SUMMARY

- Languages: Python, Java, MATLAB
- Tools: SolidWorks, Fusion 360, MATLAB, Autodesk CFD, Open Rocket, OpenVSP/VSPaero
- Skills: CFD simulation, aerodynamic analysis, CAD modeling, wind tunnel testing and analysis

EXPERIENCE

- Robotics For Space Exploration - Aerial** University of Toronto
• *Mechanical Sub-team Recruit* *Sept 2025 - Present*
 - Designed and ran Autodesk CFD simulations on the rocket nose cone. Iterated geometry on CFD and rocket simulation software.
 - 3D modelled SG90 servo mount to enable wind tunnel testing on various airfoil at different angles of attack
 - Designed a glider configuration with an optimized fuselage for low reynolds number flight. Iterated on 20+ airfoil geometries on OpenVSP.
- STEM Fellowship REO** University of Calgary
• *Student Lab Intern* *March 2025*
 - Selected out of a competitive group of ≈ 150 applicants. Worked with grad students and PIs on the extent to which seawater and the carbon cycle can remove CO2 from the atmosphere.
 - Used IC, OES, and SEM machines to decipher and analyze chemical samples. Presented findings to an audience of ≈ 60 .
- CWSF Finalist** Youth Science Canada
• *Finalist* *December 2024 - Jun 2025*
 - 3D modeled various airfoils for testing using CAD. 2nd Best Senior Project Award at CYSF. Gold Medal recipient at CYSF.
 - 1 of 15 projects selected to compete in CWSF out of ≈ 600 total projects. Bronze Medal recipient at CWSF out of ≈ 350 projects.
- STEM Fellowship** Henry Wise Wood High School
• *Senior Vice President | Treasurer* *Oct 2024 - Jun 2025*
 - **Senior VP (2024-2025):** Recruited 100+ members, oversaw club operations, and presented a science demo to 1000+ students at a pep rally.
 - **Treasurer (2023-2024):** Led members through various STEM activities. Maintained member engagement
- Brother's Taste of Asia**
• *Food Runner | Waiter* *Oct 2024 - Jun 2025*
 - Worked in a team and communicated effectively throughout. Ensured that food was up to standards and packed correctly for takeout.
 - Took phone calls, waited tables, and prepared ingredients. Trained new hires and led them throughout the restaurant.

PROJECTS

- **Vortex Generator Research Project:** Conducted a research project optimizing vortex generators (VGs) for boundary layer separation mitigation on a NACA 2412 airfoil. Identified optimal VG long gap width and placement using a homemade flow visualization wind tunnel, to significantly increase stall angle and improve aerodynamic performance. Airfoil was mathematically designed on Desmos before importing into CAD software to be edited. Presented findings at CYSF and CWSF. Link to the paper: [found here](#).
- **Citation Machine:** Citation machine with functional UI written in Java. Able to generate citations in APA, MLA, and Chicago. Works for PostMedia, TorStar, and Canadian Encyclopedia branded websites. Involves web scraping through Java. GitHub for the project: [found here](#).

VOLUNTEER EXPERIENCE

- **Math and Science Tutor** Nov 2023 - Jun 2024
• *Henry Wise Wood* *Tutored a set of students through IB and non-IB courses, including Math and Physics.*
- **Food Preparer** Sept 2022 - Jan 2025
• *Brown Bagging for Calgary's Kids* *Prepared over 2000 lunches for elementary students across Calgary.*